

## WESTERN ISLANDS, ON, Canada

WMO: 714380

Lat: 45.036N

Lon: 80.358W

Elev: 627

StdP: 14.37

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | -2.3       | 3.0 | -9.3                          | 3.4 | -0.3 | -5.0 | 4.3 | 4.2  | 47.6                  | 25.2 | 43.0 | 26.9 | 17.0                  | 0    | 0.839 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 5.8                    | 76.5            | 68.7 | 74.5 | 67.9 | 72.7 | 67.0 | 71.2                | 74.1 | 69.8 | 72.8 | 68.4 | 71.6 | 12.6                 | 170  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 70.2                            | 113.8 | 73.1 | 68.7 | 107.9 | 71.9 | 67.2 | 102.4 | 70.8 | 35.5          | 74.3 | 34.3 | 73.0 | 33.1 | 71.5 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 42.0          | 36.9 | 33.6 | DB                         | -8.1 | 81.8               | 5.0 | 2.3                                                | -11.7 | 83.5       | -14.6 | 84.8       | -17.5 | 86.1       | -21.1 | 87.8 | (4) |
| (5)               |      |      | WB                         | -8.6 | 73.3               | 4.6 | 1.9                                                | -11.9 | 74.7       | -14.7 | 75.8       | -17.3 | 76.8       | -20.6 | 78.2 | (5) |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 44.7   | 22.9  | 22.5  | 28.2  | 38.3  | 49.2  | 59.5  | 65.4  | 66.5  | 61.3  | 51.3  | 39.4  | 29.8 |
|                                                                               | DBStd       | 17.59  | 10.43 | 9.71  | 8.83  | 6.76  | 7.30  | 6.53  | 5.56  | 5.27  | 6.37  | 6.94  | 8.66  | 9.38 |
|                                                                               | HDD50       | 3780   | 839   | 771   | 676   | 356   | 105   | 7     | 0     | 0     | 3     | 68    | 329   | 626  |
|                                                                               | HDD65       | 7651   | 1304  | 1191  | 1140  | 801   | 491   | 188   | 63    | 45    | 144   | 426   | 767   | 1091 |
|                                                                               | CDD50       | 1826   | 0     | 0     | 1     | 5     | 81    | 290   | 477   | 511   | 341   | 108   | 12    | 0    |
|                                                                               | CDD65       | 226    | 0     | 0     | 0     | 0     | 2     | 22    | 76    | 92    | 33    | 1     | 0     | 0    |
|                                                                               | CDH74       | 222    | 0     | 0     | 0     | 0     | 7     | 31    | 80    | 79    | 25    | 0     | 0     | 0    |
|                                                                               | CDH80       | 7      | 0     | 0     | 0     | 0     | 0     | 1     | 4     | 2     | 0     | 0     | 0     | 0    |
| Wind                                                                          | WSAvg       | 17.5   | 21.8  | 19.4  | 17.4  | 16.6  | 15.3  | 13.5  | 13.6  | 14.0  | 15.5  | 19.8  | 21.3  | 21.4 |
| Precipitation                                                                 | PrecAvg     | 39.0   | 2.9   | 2.1   | 2.2   | 3.2   | 3.5   | 3.3   | 3.3   | 3.7   | 3.9   | 4.3   | 3.5   | 3.0  |
|                                                                               | PrecMax     | 46.1   | 4.7   | 4.4   | 5.5   | 5.8   | 6.1   | 5.6   | 5.4   | 6.6   | 7.6   | 7.0   | 5.4   | 6.1  |
|                                                                               | PrecMin     | 31.9   | 1.5   | 0.8   | 0.3   | 1.2   | 1.3   | 1.1   | 1.5   | 1.9   | 1.5   | 1.9   | 1.6   | 1.3  |
|                                                                               | PrecStd     | 3.8    | 0.8   | 0.9   | 1.1   | 1.1   | 1.2   | 1.1   | 1.1   | 1.1   | 1.4   | 1.4   | 1.0   | 0.9  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 46.1  | 43.7  | 55.1  | 61.8  | 73.7  | 77.7  | 79.2  | 79.1  | 77.3  | 69.0  | 58.3  | 51.7 |
|                                                                               |             | MCWB   | 44.3  | 41.0  | 48.8  | 55.7  | 62.1  | 66.3  | 69.5  | 70.0  | 68.9  | 62.1  | 53.8  | 49.8 |
|                                                                               | 2%          | DB     | 41.7  | 40.0  | 47.3  | 55.4  | 68.8  | 73.8  | 76.2  | 76.1  | 73.9  | 65.6  | 54.9  | 47.2 |
|                                                                               |             | MCWB   | 39.8  | 37.7  | 43.0  | 48.3  | 58.0  | 64.9  | 69.0  | 69.6  | 67.8  | 60.2  | 51.0  | 45.0 |
|                                                                               | 5%          | DB     | 38.6  | 37.2  | 41.5  | 50.9  | 64.4  | 70.9  | 74.0  | 74.1  | 71.7  | 62.9  | 53.4  | 43.9 |
|                                                                               |             | MCWB   | 36.8  | 34.9  | 38.0  | 44.4  | 56.3  | 63.0  | 68.0  | 68.5  | 67.2  | 58.4  | 48.8  | 41.1 |
|                                                                               | 10%         | DB     | 36.2  | 34.9  | 37.8  | 47.0  | 60.2  | 68.4  | 72.1  | 72.4  | 69.6  | 60.5  | 52.5  | 41.7 |
|                                                                               |             | MCWB   | 34.4  | 33.1  | 34.8  | 41.5  | 53.4  | 62.6  | 66.7  | 67.4  | 65.4  | 56.4  | 48.6  | 38.9 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 44.8  | 42.3  | 50.5  | 58.4  | 65.0  | 70.2  | 73.2  | 73.3  | 71.4  | 63.9  | 55.4  | 50.6 |
|                                                                               |             | MCDB   | 46.3  | 42.9  | 54.5  | 61.1  | 71.6  | 73.9  | 75.8  | 76.2  | 74.8  | 66.9  | 57.0  | 51.9 |
|                                                                               | 2%          | WB     | 40.1  | 38.0  | 42.9  | 50.0  | 61.2  | 67.6  | 71.0  | 71.5  | 69.7  | 61.3  | 52.2  | 45.3 |
|                                                                               |             | MCDB   | 41.1  | 39.7  | 46.9  | 52.8  | 65.6  | 71.1  | 73.9  | 74.1  | 72.4  | 64.6  | 54.3  | 47.0 |
|                                                                               | 5%          | WB     | 36.9  | 35.4  | 38.4  | 45.7  | 57.6  | 65.9  | 69.5  | 69.9  | 68.1  | 59.1  | 49.3  | 41.5 |
|                                                                               |             | MCDB   | 38.5  | 36.7  | 40.9  | 49.5  | 62.7  | 69.3  | 72.5  | 73.0  | 71.2  | 61.9  | 52.7  | 43.5 |
|                                                                               | 10%         | WB     | 34.6  | 33.1  | 35.1  | 42.0  | 54.6  | 63.9  | 67.9  | 68.4  | 66.0  | 56.9  | 48.0  | 39.0 |
|                                                                               |             | MCDB   | 35.9  | 34.8  | 37.1  | 46.1  | 58.9  | 67.2  | 71.2  | 71.5  | 69.3  | 60.2  | 52.2  | 41.3 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 11.0  | 10.5  | 9.8   | 10.0  | 12.4  | 10.8  | 6.7   | 5.8   | 7.0   | 7.3   | 7.6   | 8.7  |
|                                                                               |             | MCDBR  | 11.9  | 10.0  | 13.4  | 17.5  | 18.2  | 13.3  | 8.9   | 7.8   | 8.5   | 9.9   | 7.6   | 9.1  |
|                                                                               |             | MCWBR  | 11.3  | 9.4   | 10.2  | 12.0  | 11.7  | 8.0   | 6.0   | 6.0   | 6.9   | 7.8   | 7.2   | 8.8  |
|                                                                               | 5% WB       | MCDBR  | 12.0  | 10.6  | 13.6  | 17.0  | 17.0  | 11.5  | 8.2   | 7.3   | 8.6   | 9.2   | 9.6   | 8.6  |
|                                                                               |             | MCWBR  | 11.7  | 10.6  | 11.4  | 13.9  | 12.5  | 8.3   | 6.8   | 6.6   | 7.6   | 8.5   | 9.6   | 8.9  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.287  | 0.293 | 0.312 | 0.350 | 0.377 | 0.402 | 0.406 | 0.402 | 0.367 | 0.348 | 0.315 | 0.286 |      |
|                                                                               | taud        | 2.436  | 2.364 | 2.376 | 2.366 | 2.389 | 2.340 | 2.363 | 2.363 | 2.446 | 2.461 | 2.502 | 2.496 |      |
|                                                                               | Ebn at Noon | 265    | 284   | 293   | 289   | 283   | 275   | 272   | 269   | 271   | 260   | 252   | 254   |      |
|                                                                               | Edh at Noon | 23     | 29    | 33    | 36    | 37    | 39    | 38    | 36    | 31    | 26    | 21    | 19    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 383    | 676   | 1116  | 1452  | 1733  | 1897  | 1921  | 1634  | 1262  | 737   | 409   | 272   |      |
|                                                                               | RadStd      | 46     | 76    | 113   | 131   | 165   | 136   | 103   | 88    | 120   | 83    | 63    | 32    |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (37 neighbors) |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page